

Reviewer Pack — Schatten p-Norm Lower Bound for Disjointness Communication C...

Entry 835e60ff99be · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-10 05:57:10 UTC

Schatten p-Norm Lower Bound for Disjointness Communication Complexity

Entry ID: 835e60ff99be **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-10 05:57:10 UTC

1. Conjecture

Field A (mathematical branch): Noncommutative L^p Geometry **Field B** (complexity object): Communication Complexity of DISJOINTNESS

Statement:

For any $n \times n$ matrix M representing a DISJOINTNESS instance, the Schatten p -norm $\|M\|_p \geq \Omega(\sqrt{n})$ for $p=2$. This bound is tight for the standard DISJOINTNESS matrix and holds for all matrices with rank $\leq \log n$.

Rationale (proposer's reasoning):

Noncommutative L^p norms capture operator-theoretic structure of tensor products, which may encode the entanglement required for communication. The Schatten $p=2$ norm (Frobenius) relates to the Hilbert-Schmidt energy of the matrix, potentially mirroring the information-theoretic cost of coordination.

Taxonomy category: COMM_DISJ (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): c10db6e4d5ae31bf

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.95	SAFE	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
import math
from itertools import combinations

def run_trial(seed: int) -> dict:
    n = 40
    random.seed(seed)

    def generate_disjointness_matrix(n):
        M = [[0] * n for _ in range(n)]
        pairs = list(combinations(range(n), 2))
        for i, j in random.sample(pairs, min(len(pairs), n // 2)):
            M[i][j] = M[j][i] = 1
        return M

    def matrix_multiplication(A, B):
        m, k, n = len(A), len(B[0]), len(B)
        C = [[0] * n for _ in range(m)]
        for i in range(m):
            for j in range(n):
                for l in range(k):
                    C[i][j] += A[i][l] * B[l][j]
        return C

    def transpose(A):
        m, n = len(A), len(A[0])
        T = [[0] * m for _ in range(n)]
        for i in range(m):
            for j in range(n):
                T[j][i] = A[i][j]
        return T

    def svd(M):
        U = M
        V = transpose(U)
        S = [sum(x**2 for x in row) ** 0.5 for row in U]
        return U, S, V

    def frobenius_norm(M):
```

```

    m, n = len(M), len(M[0])
    norm = 0
    for i in range(m):
        for j in range(n):
            norm += M[i][j] ** 2
    return norm ** 0.5

M = generate_disjointness_matrix(n)
frob_norm = frobenius_norm(M)
rank = sum(1 for s in svd(M)[1] if abs(s) > 1e-6)

metric_name = "Frobenius Norm"
metric_value = frob_norm
instances_tested = 1
conjecture_holds = frob_norm >= 0.9 * math.sqrt(n)
counterexample = "" if conjecture_holds else f"Frobenius norm {frob_norm} < 0.9√{n}"

return {
    "metric_name": metric_name,
    "metric_value": metric_value,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i - 1 for i in range(5, 30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = (sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results)) ** 0.5
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"Frobenius norm < 0.9√{n}\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

unterexample': ''
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, 'conjecture_holds': True}

```

```

TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
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TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
RESULT: SUPPORTED mean=6.324555320336759 std=0.0 support_fraction=1.0

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

n too small: only 1 instance tested, which is insufficient to establish scaling behavior. The metric value $6.3245 \approx \sqrt{40}$ suggests the test case may have been constructed to exactly meet the lower bound, not demonstrating asymptotic growth. The conjecture claims $\Omega(\sqrt{n})$ but no data shows how the metric scales with n.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Single instance tested (n=40) meets the claimed lower bound exactly, but insufficient to establish asymptotic scaling. The metric value $6.3245 \approx \sqrt{40}$ suggests a constructed test case rather than empirical validation of $\Omega(\sqrt{n})$ for growing n. | next: Test with larger n (e.g., n=1000) to observe scaling behavior and verify the lower bound holds asymptotically

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	112415
2	propose	ollama_remote	qwen3:8b	0	0	101235
3	propose	ollama_remote	qwen3:8b	0	0	111693
4	propose	ollama_remote	qwen3:8b	0	0	58200
5	preregistration	ollama_remote	qwen3:8b	0	0	24104
6	novelty	ollama_remote	qwen3:8b	0	0	20771
7	novelty	ollama_remote	qwen3:8b	0	0	14155
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14499
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10291
10	critic	ollama_remote	qwen3:8b	0	0	29007
11	judge	ollama_remote	qwen3:8b	0	0	18407

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 514778 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in [research/audit/835e60ff99be.jsonl](#))

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/835e60ff99be.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/835e60ff99be.tar.gz` (if generated)